



UNIVERSITY OF KELANIYA
DEPARTMENT OF MATHEMATICS
ACADEMIC YEAR 2022/2023
BACHELOR OF SCIENCE (GENERAL) DEGREE
PMAT 21263 - LINEAR ALGEBRA

Semester I

Tutorial 06

Due date : 24.06.2024 at 4.00 p.m.

Submit only 01, 02, 03)b), 04, 05, 06)c), 06)d)

1. (a) What does it mean to say that the set of vectors $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ in a vector space V is a basis for V ?
Show that if S is a basis of V , then any vector of V can be uniquely written as a linear combination of the vectors in S .
- (b) Let A be any $n \times n$ matrix, and $\mathbf{v}_1$ and $\mathbf{v}_2$ be two non-zero vectors. Suppose that $A\mathbf{v}_1 = 2\mathbf{v}_1$ and $A\mathbf{v}_2 = 5\mathbf{v}_2$. Prove that the set $\{\mathbf{v}_1, \mathbf{v}_2\}$ is linearly independent.
- (c) Let $W = \{\mathbf{x} \in \mathbb{R}^3 : x - 2y + z = 0\}$.
Find a basis for W and determine the dimension of W .
2. (a) Which of the following are subspaces of the given vector spaces V ?
 - i. $S_1 = \left\{ \begin{bmatrix} 2a & b \\ 3a+b & 3b \end{bmatrix} : a, b \in \mathbb{R} \right\}, V = M_{2 \times 2}$
 - ii. $S_2 = \{(x, y, z) : x^2 = z^2\}, V = \mathbb{R}^3$
 - iii. $S_3 = \{(x, y, z) : x, y, z \geq 0\}, V = \mathbb{R}^3$
- (b) Let $\mathbf{u} = (-3, 5)$ and $\mathbf{v} = (7, 2)$. Show that $\{\mathbf{u}, \mathbf{v}\}$ spans $\mathbb{R}^2$
- (c) For what value of k , the polynomial $x^2 - 2x + k \in \mathcal{P}_2$ is a linear combination of polynomials $3x^2 - 2$ and $2x^2 - x - 5$.
- (d) Let U and W be to subspaces of vector space V . Show that $U \cap W$ is also a subspace of V .
3. (a) Find the conditions on a, b and c so that $\mathbf{v} = (a, b, c) \in \mathbb{R}^3$ belongs to $\text{span}(\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3)$, where $\mathbf{u}_1 = (1, 2, 0)$, $\mathbf{u}_2 = (-1, 1, 2)$ and $\mathbf{u}_3 = (3, 0, -4)$.
- (b) i. Let U and W be two subspaces of the vector space V . The smallest subspace containing U and W is given by $U + W$. Show that, in usual notation, $U + W = \text{span}(U, W)$.

ii. Let $B_1 = \left\{ \begin{pmatrix} 1 \\ -1 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} -1 \\ 2 \\ -1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ -1 \\ 0 \\ 0 \end{pmatrix} \right\}$

and

$$B_2 = \left\{ \begin{pmatrix} 0 \\ 1 \\ -1 \\ 1 \end{pmatrix}, \begin{pmatrix} -1 \\ 1 \\ 1 \\ 1 \end{pmatrix} \right\}$$

be two bases for subspaces U and W of $\mathbb{R}^4$, respectively. Find bases for $U + W$ and $U \cap W$. Deduce that, in the usual notations,

$$\dim(U + W) = \dim(U) + \dim(W) - \dim(U \cap W)$$

4. (a) Let V be a vector space of dimension n . If $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ is linearly independent. Show that it is a basis for V .
- (b) Assume that, if v is a vector in V that is not in $\text{span}(S)$, then $T = \{\mathbf{v}, \mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n, \mathbf{v}\}$ is linearly independent. Let $S_1 = \{v_1, v_2, \dots, v_r\}$ be a linearly independent set of vectors in an n -dimensional vector space V with $r < n$. Then, show that S_1 can be extended to a basis for V .
- (c) Show that subset $S_1 = \{(1, 0, 1, 0)^T, (-1, 1, -1, 0)^T, (1, -1, 0, 0)^T\}$ is linearly independent.
- (d) Extend the linearly independent set S to a basis B for $\mathbb{R}^4$.
- (e) Find the coordinate vector of $(1, -3, 2, 0)^T$ with respect to the basis B .
5. (a) Let V be a vector space, $\mathbf{u} \in V$ and k be a scalar. Show that
 - i. $0\mathbf{u} = \mathbf{0}$
 - ii. $(-1)\mathbf{u} = -\mathbf{u}$
- (b) Let U and W are subspaces of a vector space V . Show that $U + W$ defined by $U + W = \{\mathbf{u} + \mathbf{w} : \mathbf{u} \in U, \mathbf{w} \in W\}$ is a subspace of V .
- (c) Let $W = \{(a, b, c) : a^2 + b^2 + c^2 \leq 1\}$. Show that W is not a subspace of $\mathbb{R}^3$.

(d) Find the dimension of the subspace $H = \begin{cases} a - 3b + 6c \\ 5a + 4d \\ b - 2c - d \\ 5d \end{cases} \quad a, b, c, d \in \mathbb{R}.$

6. (a) Determine whether the vectors $\mathbf{v}_1 = (1, 1, 2)$, $\mathbf{v}_2 = (1, 0, 1)$, $\mathbf{v}_3 = (2, 1, 3)$ span the vector space $\mathbb{R}^3$.
- (b) Find the values of k so that the vectors $(3 - k, -1, 0)$, $(-1, 2 - k, -1)$, $(0, -1, 3 - k)$ span a 2-dimensional subspace.
- (c) Let $W = \{(2b - 3a, a, b) : a, b \in \mathbb{R}\}$. Find the basis for W and determine the dimension.
- (d) Determine whether the polynomials $\mathcal{P}(t) = t^3 - 3t^2 + 5t + 1$, $\mathcal{Q}(t) = t^3 - t^2 + 8t + 2$, $\mathcal{R}(t) = 2t^3 - 4t^2 + 9t + 5$ in the vector space of all polynomials of degree less than or equal to 3, are linearly independent.